

A glass-box process simulator

Philosophy, numerics, and the solubility data of Prof. Fátima Farelo

Chemical Engineering Department (DEQ)

Instituto Superior Técnico — Técnico Lisboa

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Where we are going (30 minutes)

- ① A short history of process simulators — and the closed-source turn.
- ② The philosophy and architecture of **Choupo** (glass-box, open).
- ③ The electrolyte *model* and the *numerical method*, and how they are implemented.
- ④ **Prediction**: Farelo's $\text{NaCl} + \text{NH}_4\text{Cl}$ salting-out.
- ⑤ **Recalibration**: Farelo's LiCl , where the model first fails.
- ⑥ How a measurement *enters the compound* — the data flow.

1960s–70s: flowsheeting is born — and it is open

- **1966** — Monsanto's *FLOWTRAN*; PACER, CHESS: the first sequential-modular flowsheet programs. Shared between universities and industry, source listings circulated.
- **1976** — the *ASPEN* project at **MIT** (Advanced System for Process Engineering), led by L. B. Evans, funded by the US DOE. A **public, academic** effort to build a rigorous open simulator.

The irony that frames this talk

The archetype of today's *closed* simulator began as an *open*, publicly-funded, university project.

- **1981** — AspenTech founded; ASPEN → *Aspen Plus*, proprietary, licensed.
- HYSYS (Hyprotech), PRO/II (SimSci), ChemCAD, ProSim — a proprietary ecosystem.
- Consolidation: AspenTech acquires Hyprotech (2002); the market becomes a handful of expensive, closed giants.

What “closed” costs the engineer

You cannot *see* the equation, *question* the thermodynamic model, *audit* the convergence, or *extend* it to your problem. The simulator is a black box; the engineer is a button-pusher who feeds inputs and trusts the output.

“Professor(a), eu penso que. . .”

*“Você não pensa! Para pensar estamos cá nós!
Você só faz o que lhe mandamos. . .”*

— antigo docente do DEQ

A black-box simulator treats the engineer exactly like that: *you do not think — you do what you are told.*

Choupo is the opposite premise.

The premise. Every equation, every Newton iteration, every K -value, every activity coefficient is visible — in the *source* and in the *run-time output*. Nothing is hidden, autogenerated, or magic.

- C++17, no heavy dependencies; hand-rolled Newton, Gauss, RK4.
- GPL-3.0 — free for academia, research, industry.
- Pedagogical *and* a research differentiator (membranes, NF/RO, electrolytes, exotic domains).

The credo

No silent crutch. Every solver aid is explicit and announced.

All models are wrong, some useful — so the model's *limits* are shown, not hidden.

File-first. A case is plain-text dictionaries on disk (no GUI lock-in, no binary project file). Edited with any editor — increasingly with an LLM. **Explicit registries.** Every model (unit op, activity, EoS, reactor) is registered by one line in `main.cpp` — no auto-registration, no static-init magic. The student sees where types come from.

Three layers

OuterDriver — sensitivity, optimisation, parameter estimation.

Simulator core — Flowsheet → streams + KPIs (one pass).

PostProcessor — sizing, costing, reports.

The data catalogue

`data/standards/`: components, ions, Pitzer pairs, minerals — each value with *per-value provenance* (cite the *primary* source).

The electrolyte model: Pitzer–Harvie–Møller–Weare

A per-ion activity coefficient on the molality scale, referenced to infinite dilution in water:

$$\ln \gamma_i = \underbrace{z_i^2 f^\gamma(I)}_{\text{Debye-Hückel}} + \underbrace{\sum_a m_a (2B_{ia} + ZC_{ia})}_{\text{binary virials}} + \underbrace{\sum m \Phi, \psi}_{\text{ternary mixing}} + \underbrace{E\theta(I)}_{\text{higher-order electrostatic}}$$

- Binary $\beta^{(0)}, \beta^{(1)}, C^\phi$ per (cation, anion) pair.
- Ternary θ_{ij} (like-sign) and ψ_{ijk} (triplets).
- $E\theta(I)$ — the unsymmetrical mixing term most codes hard-wire on; Choupo computes it from the $J(x)$ kernel *and announces it*.

The skeptic's gate (glass-box)

`PitzerHMW::verify` reduces the multi-ion model to every single salt in the catalogue and checks it against the closed single-salt kernel: max deviation 1.6×10^{-14} (γ) and 1.9×10^{-15} (ϕ).

The numerical method: the speciation solver

A chemical-equilibrium network solved as $F(x) = 0$:

- Unknowns $x_j = \ln m_j$ (positivity built in) + $\ln a_H$ when pH is solved by electroneutrality $\sum_i z_i m_i = 0$.
- **Augmented Newton** in $\ln$ -space: one mass-action residual per species, one mole balance per master, the charge-balance row.
- Wrapped in an **ionic-strength fixed point**: freeze $\gamma_i(I)$ and a_w , solve Newton, refresh I , repeat to self-consistency.
- Minerals: an **active-set** loop — admit the most supersaturated, evict any with negative amount, re-solve, until complementarity ($SI \leq 0$ with $n = 0$, or $SI = 0$ with $n > 0$).

Announced, not hidden

At `verbosity` ≥ 3 the solver prints the active parameters, every γ pass, every Newton residual, every mineral admitted. You can *think*.

The solvent's non-ideality: the osmotic coefficient

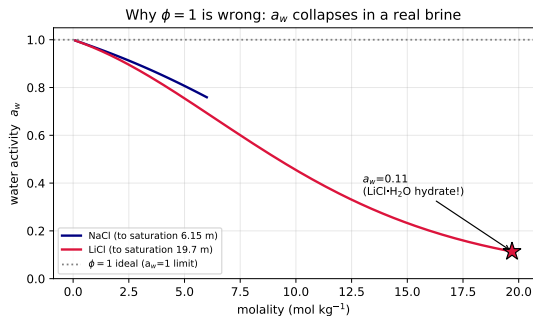
The water activity is *not* an afterthought:

$$\ln a_w = -M_w \phi \sum_i m_i.$$

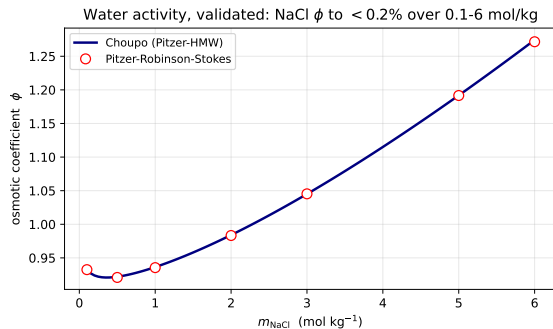
ϕ is the osmotic coefficient, built from the *same* virials as the γ_i (Gibbs–Duhem forces it).

The common shortcut $\phi = 1$ is exactly the error that wrecks a concentrated-brine or osmosis calculation — because a_w *collapses*.

At LiCl saturation $a_w = 0.11$ — the hydrate equilibrium lives or dies by it. (Implemented this in Choupo's PitzerHMW; the interface had reserved the slot for it.)



Validation of the water activity



- NaCl ϕ vs Pitzer–Robinson–Stokes, 0.1–6 mol/kg: < 0.2%.
- Seawater $a_w = 0.981$ (measured 0.980–0.981).
- Single-salt ϕ oracle to 10^{-15} .
- **Independent adversarial audit** (four expert reviewers, term by term vs Pitzer 1991 / HMW 1984): zero defects.

This is what “glass-box” buys: the validation is *in the repository*, reproducible, and was *tried to be broken* on purpose.

Farelo, Fernandes & Avelino, *Solubilities for Six Ternary Systems... at $T = (298\text{--}333) \text{ K}$* , J. Chem. Eng. Data **50** (2005) 1470. DOI 10.1021/je050111j.

- Six chloride ternaries; saturation molalities to $\pm 0.0007 \text{ mol/kg}$.
- Pure-salt anchors (298 K): NaCl 6.15, NH_4Cl 7.35, LiCl 19.7 mol/kg.
- NaCl+ NH_4Cl : simple eutonic, no double salt — an *experimental* fact the model assumes.

The test

Calibrate the model to *only* the two pure solubilities. Then *predict* the mixed-system saturation, and compare to her measured points.

The measured quantity: the composition of the *saturated solution* — the molality of *each* salt in the liquid in equilibrium with the solid phase(s) — mapped from one pure-salt edge to the other, on isotherms at $T = (298, 313, 333)$ K (some series to 340 K).

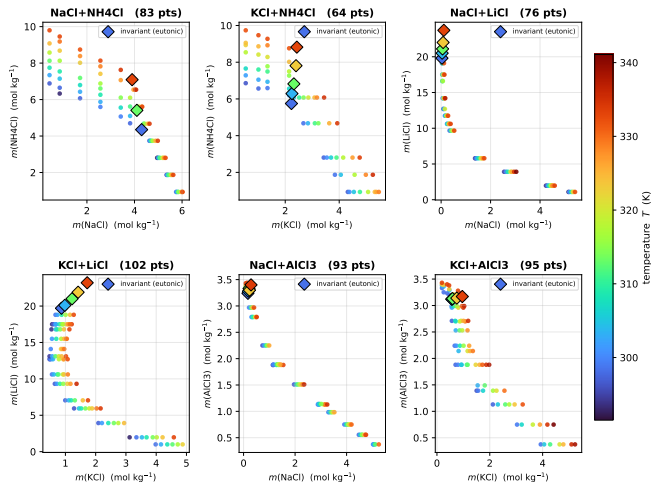
ternary	T/K	measured molality range (mol/kg)		solid phase(s)
NaCl+NH ₄ Cl	298–333	NaCl 3.9–6.15	NH ₄ Cl 4.35–7.35	NaCl + NH ₄ Cl
KCl+NH ₄ Cl	298–333	KCl 2.23–2.43	NH ₄ Cl 5.75–8.82	KCl + NH ₄ Cl
NaCl+LiCl	298–333	NaCl 0.03–6.15	LiCl 19.7–23.5	LiCl·H₂O
KCl+LiCl	298–333	KCl 0.87–4.78	LiCl 19.7–23.5	LiCl·H₂O
NaCl+AlCl ₃	298–338	NaCl 0.17–6.15	AlCl ₃ 3.24–3.45	AlCl₃·6H₂O
KCl+AlCl ₃	298–338	KCl 0.57–4.78	AlCl ₃ 3.12–3.45	AlCl₃·6H₂O

How: visual polythermal (synthetic) method, sealed cell ± 0.03 K (chloride fields); isothermal equilibration + AgNO₃ potentiometric Cl[−] titration ($\pm 0.8\%$) + AAS (Li/Al systems). Saturation molalities to ± 0.0007 mol/kg.

And: the *solid phase itself* — eutonic (no double salt) vs hydrate vs hydrolysis — an experimental finding the model cannot guess.

All 553 measured points

All of Farelo's saturation data (J. Chem. Eng. Data 50, 2005, 1470): six ternary chloride systems, 553 measured points



Every saturated composition from the paper's twelve tables — six ternary systems, coloured by temperature.

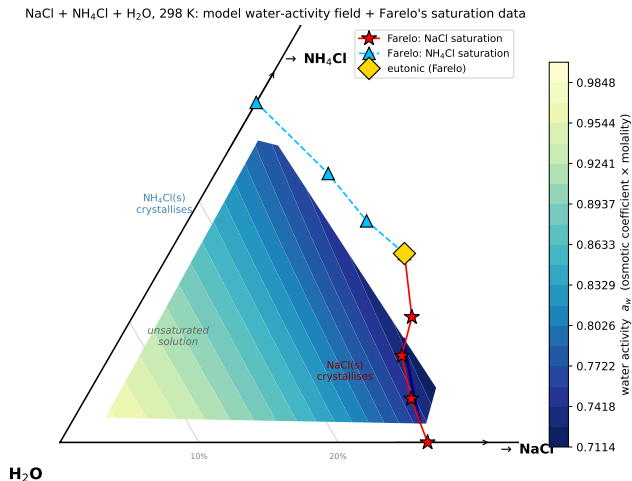
The plan: predict, regress — and *where* it is stored

One thermodynamic model throughout: **Pitzer–Harvie–Møller–Weare** (γ_i + the osmotic a_w) for the liquid, K_{sp} for each solid.

systems	what we do	parameters <i>obtained</i>	saved in
NaCl/KCl + NH ₄ Cl	predict the ternary; fit one K_{sp}	NH ₄ –Cl β^0, β^1, C^ϕ sal-ammoniac $\log K_{sp}$	pairs.dat minerals.dat
NaCl/KCl + LiCl	regress (refit) to reach saturation	Li–Cl β^0, β^1, C^ϕ (refit) LiCl·H ₂ O $\log K_{sp}$ Li ⁺ $\Delta_f H_{aq}^\circ$	pairs.dat minerals.dat ions.dat
NaCl/KCl + AlCl ₃	defer — needs an Al ³⁺ hydrolysis model	(none yet)	—

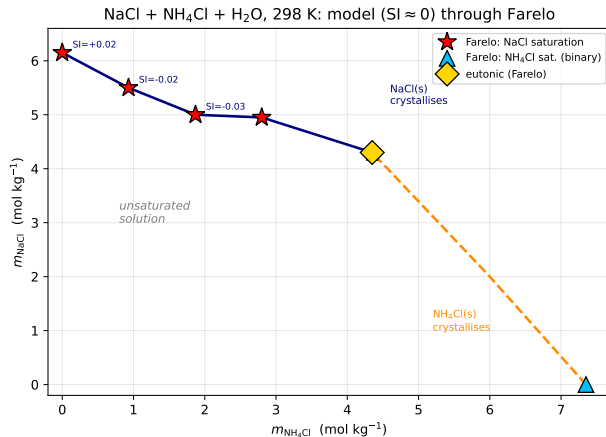
Every obtained parameter becomes a *permanent, sourced* line in data/standards/electrolyte/ — cited to Farelo, read by name by the engine, pinned by a validation case. (The exact pairs.dat entry: later.)

The salting-out, in the ternary diagram



The model's water-activity field (colour) with Farelo's two measured saturation branches and the eutonic. a_w falls 0.98 $\rightarrow$ 0.71 toward saturation — the osmotic drive for crystallisation

The common-ion salting-out, quantified



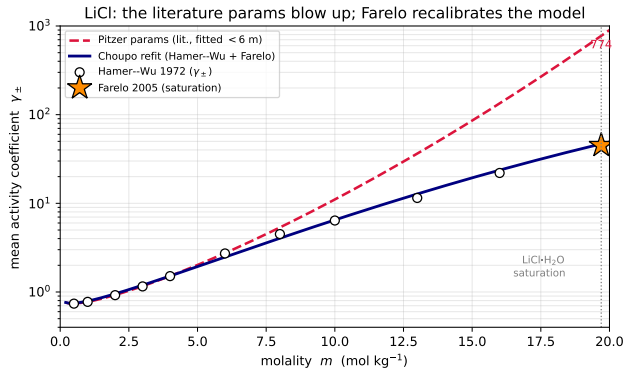
Adding NH₄Cl drives NaCl out of solution (shared Cl⁻). The model's NaCl saturation index at each of Farello's *measured* points:

point	SI _{halite}
NaCl 6.15	+0.017
+ NH ₄ Cl 0.93	-0.022
+ NH ₄ Cl 1.87	-0.028
NH ₄ Cl 7.35	-0.000

± 0.03 log units.

The IST simulator reproduces the IST measurements.

LiCl: where the model first *fails* — then is fixed



LiCl saturates at ~ 20 mol/kg. The literature parameters (fitted <6 m) give $\gamma_{\pm} = 774$ vs ~ 45 measured. **Don't declare defeat** —

use the measurement. Refit β , C^{ϕ} to Hamer–Wu $\gamma_{\pm}$ and Farelo's saturation. A *negative* C^{ϕ} tames the blow-up; the model now tracks LiCl to $\sim 11\%$ to saturation. And the LiCl·H₂O hydrate K_{sp} closes *through* the osmotic $a_w = 0.11$ — the two pieces of this talk meeting.

The data flow — measurement → catalogue → engine

- 1 **Measurement** (Farelo 2005): LiCl saturation 19.7 mol/kg; NH₄Cl 7.35.
- 2 **Curation** (a human act): fit / calibrate, with the *primary* source cited per value. The engine **refuses** to write into data/standards/ — adding data is deliberate.

what actually lands in electrolyte/pairs.dat

```
{ cation Li; anion Cl; beta0 0.17; beta1 0.278; Cphi -0.0026;  
  source "REFIT to Hamer & Wu 1972 gamma_pm + Farelo (JCED 50, 2005, 1470)  
  saturation, valid to ~19 mol/kg; supersedes the Appelo 2014 low-m fit"; }
```

- 3 **Engine** reads it *by name*; PitzerHMW uses it; the run *announces* the active parameters.
- 4 **Validation case** pins the result to the measurement (a golden master).

The whole chain is open, sourced, and re-runnable — the antithesis of the black box.

Where the model is valid

it *earns trust* by reproducing the measurement (NaCl+NH₄Cl, $S_I \approx 0$). The water activity becomes a *prediction*, not a guess.

Where it fails

the measurement is *irreplaceable* (LiCl at saturation) — and it *calibrates* the model so it can be extended *honestly*.

A model that only reproduces what it was fitted to is *consistent*; the experiment that grounds it is what makes it *useful*.

And in a glass box, the engineer can finally answer the old professor:

“Sim — e eu posso ver, e perguntar, e pensar.”

- Simulators began *open* and turned *closed*; Choupo turns back — glass-box, open, every equation visible and announced.
- A rigorous Pitzer–HMW electrolyte model and speciation solver, validated and *adversarially audited*.
- Farelo's NaCl+NH₄Cl: **predicted** to ± 0.03 log units.
- Farelo's LiCl: **recalibrated** the standard parameters, and the hydrate closed through the osmotic water activity.
- A respected IST colleague's dataset is now the reference curve for an open, IST-built simulator.

Obrigado — e obrigado, Professora Farelo.